Stormwater Regulation Summary: San Francisco, CA

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Combined sewer overflow reduction and Bay water quality protection.
- San Francisco Stormwater Management Requirements (SMR) administered by SFPUC.
- Applies to projects with ≥5,000 SF of new or replaced impervious surface in combined sewer areas (≥2,500 SF in separate sewer areas).
- LID-based approach: Manage the 25th percentile storm event (approximately 0.75") via on-site LID.
- Better Roofs Ordinance: 15-30% of roof area must be solar or living roof on new construction ≥2,000 SF.

2. Typical Soil Conditions

- San Francisco soils are primarily Franciscan Complex-derived — sandstone, shale, and serpentinite weathering products.
- Typical soils: Urban land-Orthents complex (fill), Candlestick-Kron complex (clay loam on steep slopes).
- Hydrologic Soil Group: Predominantly C and D — clay-rich and often shallow to bedrock.
- Western neighborhoods (Sunset, Richmond) have sandy soils (Dune sand deposits) with better infiltration (HSG A-B).
- Implication: Infiltration potential varies dramatically by neighborhood — sandy west side vs. clay east side. Many sites use fill.
- *Seismic considerations are paramount — subsurface BMP design must account for liquefaction potential.*

3. Green Roof Requirements

- Better Roofs Ordinance (2017): 15-30% of roof area on new buildings ≥2,000 SF must be solar OR living roof (green roof).
- Bio-solar (OverEasy VPV on green roof) satisfies both the solar and living roof requirements simultaneously.
- SFPUC stormwater fee credit available for green roofs that manage on-site runoff.
- Green roofs count toward the 25th percentile storm retention requirement under the SMR.
- SF has a Mediterranean climate (dry summers) — irrigation considerations differ from East Coast green roofs.
- Lightweight extensive systems (<25 psf saturated) are preferred due to seismic weight considerations.

4. TSS Requirements

- SFPUC SMR requires LID-based treatment of the 25th percentile storm — inherently addresses TSS for that volume.
- Green roofs capture and treat the design storm volume, providing effective TSS removal for captured rainfall.
- No separate numeric TSS percentage target in the SMR — compliance is achieved by meeting the LID performance standard.
- Bay Area Regional Water Quality Control Board may impose additional TMDLs for specific pollutants in some watersheds.
- *Mercury and PCB TMDLs affect some SF watersheds — verify if additional pollutant-specific treatment is needed.*

5. Nature-Based Retention Requirements

- Retain the 25th percentile, 24-hour storm event (approximately 0.75") from all project impervious surfaces.
- Retention via LID: infiltration, evapotranspiration (green roofs), or rainwater harvesting.
- CN method is used for volume calculations where applicable, though SF often uses the Rational Method for peak flows.
- Low annual rainfall (23.6") means green roofs can capture a high percentage of annual precipitation — high retention effectiveness.
- Mediterranean climate: most rainfall occurs November-March — summer maintenance differs from humid-climate green roofs.

6. Allowable Outflow Rates

- Post-development flow duration must not exceed pre-development for the range of storms up to the 25th percentile event.
- SF does not specify a flat l/s/ha rate — requirements are LID performance-based.
- For combined sewer areas, SFPUC manages downstream capacity — site-level flow control reduces CSO frequency.
- *Allowable rates depend on specific sewer shed capacity — SFPUC reviews each project.*
- For reference, managing the 0.75" event equates to approximately controlling flows up to 5-10 l/s/ha in a typical 24-hr event.